

PAPER_899: Buoyancy Reversal Sign Flip Resonance

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Abstract

Sweep of $F_{\{U,B\}}/F_U$ from 0.1 to 0.9 detecting net_factor sign changes. The critical ratio $R=0.5$ marks the buoyancy-gravity balance point where net_factor = 0. Sign flips indicate expansion↔erosion phase transitions, analogous to cosmological phase changes.

1. Core Equations

net_factor(R) = $2R - 1$
critical ratio at $R = 0.5$
sign flip = phase transition

2. Parameters

Parameter	Default	Description
ratio_min	0.1	Minimum buoyancy ratio
ratio_max	0.9	Maximum buoyancy ratio
n_points	9	Number of sweep points

3. UQFF Integration

This calculator integrates et_phonon_resonance.py from Session 208 into the CP4 pipeline as class #483.

PAPER_899 — Star Magic UQFF Framework — CVW v2.0.0